William Anderson

Senior Software Engineer

Seminole, FL | williamandresondev@outlook.com | +1 (407) 801-1287 | https://www.linkedin.com/in/william-andreson-39b88b3aa

Professional Summary

Senior **Python AI/ML Engineer** with 12+ years of experience building production platforms for **healthcare, fintech, eCommerce, logistics, cloud SaaS, and enterprise automation**, using **Python 3.8–3.12**, **FastAPI 0.95–0.110**, **Django 3.x–4.x**, **PyTorch 1.12–2.2**, **TensorFlow 2.x**, **scikit-learn 1.x**, **LangChain 0.1.x**, **PostgreSQL 12–16**, **Docker 20.x–24.x**, **Kubernetes 1.22–1.29**, and **AWS/GCP** to deliver ML pipelines, LLM workflows, prediction services, data APIs, model monitoring, legacy migrations, and secure high-availability systems that improved automation, accuracy, and release confidence across distributed teams.

Technical Skills

- **Programming Languages: Python 3.8–3.12**, SQL, Bash, JavaScript, TypeScript
- **AI/ML Frameworks: PyTorch 1.12–2.2**, **TensorFlow 2.x**, **scikit-learn 1.x**, XGBoost, LightGBM, NumPy, Pandas
- **LLM & AI Engineering: LangChain 0.1.x**, OpenAI API, RAG Pipelines, Embeddings, Vector Search, Prompt Engineering, Agent Workflows
- **Backend Frameworks: FastAPI 0.95–0.110**, **Django 3.x–4.x**, Flask 2.x, Celery 5.x, REST APIs, GraphQL
- **Data & Messaging:** PostgreSQL 12–16, MySQL 8.x, Redis 6.x–7.x, MongoDB 5.x–7.x, Kafka 2.x–3.x, RabbitMQ
- **Cloud & DevOps:** AWS, GCP, Docker 20.x–24.x, Kubernetes 1.22–1.29, Terraform 1.x, GitHub Actions, GitLab CI/CD
- **MLOps & Observability:** MLflow, Airflow 2.x, Prometheus, Grafana, OpenTelemetry, CloudWatch, Model Monitoring
- **Testing & Quality:** Pytest, unittest, Locust, Postman, SonarQube, Snyk, CI/CD Quality Gates

Professional Experience

Viaro Networks Inc.

Senior Software Engineer | *January 2021 – December 2025*

- Designed production **Python AI/ML services** with **Python 3.9–3.12**, **FastAPI 0.95–0.110**, **PyTorch 1.12–2.2**, and **PostgreSQL 13–16** for healthcare, logistics, and enterprise SaaS platforms.
- Built **LLM-powered RAG workflows** with **LangChain 0.1.x**, OpenAI APIs, vector embeddings, and document chunking pipelines to automate knowledge search and reduce manual support review time by **38%**.
- Implemented fraud and anomaly detection models with **scikit-learn 1.x**, XGBoost, feature engineering, and scheduled retraining jobs that improved suspicious transaction detection accuracy by **29%**.
- Migrated legacy **Django 2.x** API modules into **FastAPI 0.100+** microservices, replacing synchronous bottlenecks with async endpoints, typed schemas, and cleaner service boundaries.
- Fixed a high-impact prediction latency issue caused by oversized model payloads by adding model caching, batch inference, Redis lookups, and profiling around slow feature extraction steps.

- Developed **MLOps pipelines** with MLflow, Docker, GitHub Actions, and Kubernetes deployments to version models, validate metrics, and safely promote approved models into production.
- Resolved recurring data drift issues by adding distribution checks, model confidence thresholds, monitoring dashboards, and alerts that reduced silent model-quality incidents by **34%**.
- Created secure **FastAPI** authentication layers with JWT, OAuth2 flows, role-based permissions, and audit logging for AI services handling sensitive customer and operational data.
- Optimized PostgreSQL query performance by adding partial indexes, partition-aware filters, async database sessions, and pagination changes that reduced API response time by **42%**.
- Implemented new AI features for document classification, semantic search, ticket routing, and recommendation workflows while keeping fallback rules for low-confidence model outputs.
- Improved CI/CD reliability by adding Pytest coverage gates, container vulnerability scans, rollback scripts, and environment-based deployment approvals across staging and production.
- Collaborated with data scientists, product managers, and DevOps teams to translate experimental notebooks into maintainable **Python 3.11** services with clear APIs and measurable business impact.

VividCloud

Software Engineer | *January 2016 – December 2020*

- Developed cloud-based analytics and automation platforms using **Python 3.6–3.8**, **Django 1.11–3.1**, Flask 1.x, TensorFlow 1.15–2.3, and AWS services for fintech and SaaS clients.
- Built machine learning pipelines with **scikit-learn 0.20–0.23**, Pandas 0.24–1.1, NumPy, and Airflow 1.10 to support churn prediction, demand forecasting, and customer segmentation.
- Implemented a new recommendation feature using collaborative filtering, feature normalization, and scheduled batch scoring that increased personalized content engagement by **31%**.
- Fixed a production ML scoring bug caused by mismatched training and inference preprocessing by creating shared transformation modules and versioned feature contracts.
- Migrated Python 2.7 scripts into **Python 3.7–3.8**, replacing deprecated libraries, updating encoding behavior, and adding regression tests for legacy reporting workflows.
- Optimized slow Django APIs by removing N+1 queries, adding select-related joins, caching expensive aggregation results, and improving average dashboard load time by **46%**.
- Created TensorFlow 2.x migration plans for older TensorFlow 1.x models, converting session-based training code into eager execution and reusable model classes.
- Built ETL jobs with Airflow, S3, PostgreSQL, and Spark 2.4 to clean large event datasets and prepare reliable training data for analytics and prediction models.
- Resolved intermittent Celery queue failures by tuning worker concurrency, retry policies, task idempotency, and dead-letter handling for long-running data-processing jobs.
- Improved production visibility by adding CloudWatch metrics, structured Python logging, error dashboards, and alert rules that reduced incident investigation time by **35%**.
- Worked across backend, data, and infrastructure teams to deliver practical AI features while keeping deployment risk low through code reviews and staged releases.

Minimalist Moon

Software Developer | *January 2013 – December 2015*

- Built backend and data-driven web applications using **Python 2.7–3.5**, **Django 1.6–1.10**, Flask 0.10–0.11, PostgreSQL, MySQL, Redis, and Celery 3.x.
- Developed early machine learning features with **scikit-learn 0.15–0.18**, NumPy 1.8–1.11, and Pandas 0.13–0.18 for classification, scoring, and reporting workflows.
- Implemented customer behavior scoring models that helped sales and support teams prioritize high-risk accounts and improved follow-up efficiency by **27%**.
- Fixed a recurring background job duplication issue by adding Celery task locks, database-level status checks, and safer retry logic for long-running report generation.

- Migrated older PHP and manual reporting workflows into **Python Django** modules, creating reusable admin tools and reducing spreadsheet-based operations by **40%**.
- Built REST-style endpoints for internal dashboards, customer records, order tracking, and analytics exports while maintaining clean validation and database transaction handling.
- Improved SQL performance by replacing inefficient joins, adding composite indexes, and caching repeated calculations that reduced heavy report execution time by **33%**.
- Created data-cleaning scripts for messy CSV imports, duplicate records, missing values, and inconsistent timestamps before loading datasets into analytics tables.
- Supported production troubleshooting by reading logs, reproducing user issues locally, patching broken workflows, and releasing hotfixes with careful rollback notes.
- Collaborated with senior engineers to apply Python best practices, improve test coverage, and turn manual business processes into maintainable internal software.

Microsoft

Software Developer Intern | *July 2012 – December 2012*

- Supported engineering work on internal productivity and web-based workflow tools using **Python**, JavaScript, SQL-backed services, and maintainable UI enhancements appropriate to enterprise development practices of that period.
- Assisted with bug fixing, form behavior improvements, and service integration tasks, helping resolve user-facing inconsistencies and improving functional stability across shared internal modules.
- Investigated a defect where inconsistent validation between frontend input handling and backend processing caused rejected submissions, then aligned rules and reduced avoidable form failures for end users.
- Contributed to small feature delivery across data entry, search, and reporting flows while learning structured review practices, team collaboration, and disciplined release habits in a large engineering environment.
- Improved query handling and response formatting in selected internal endpoints so data-heavy pages loaded more predictably and generated fewer support questions from business users.
- Helped document issue reproduction steps, test outcomes, and implementation notes, making handoff smoother for full-time engineers and improving clarity during bug triage discussions.
- Worked within established coding standards and source control workflows to ship safe incremental changes, building strong habits around maintainability, peer feedback, and reliable delivery.
- Participated in debugging sessions involving cross-layer issues between backend responses and UI rendering, gaining practical experience in tracing failures through logs, request flow, and data state.
- Adapted quickly across backend, database, and frontend tasks, showing early flexibility with engineering fundamentals while contributing meaningfully to production-adjacent enterprise software work.

Education

Florida State University

Bachelor’s Degree in Computer Science | 2008 – 2012